COMP 640 – EXPLAINABLE NLP · GRADUATE SEMINAR

Editability–Faithfulness

Probing the Causal Grounding of Free-Text Rationales via ROME

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Talk Length: 5 minutes (excluding title + Q/A)

- Knowledge Editing
- Free-Text Rationales
- Causal Grounding

01

Motivation & Problem Setup

Explainability vs. reality

“Explainability” often stops at pretty rationales — we need to know whether they reflect the model’s actual knowledge.

EF as an intervention test

Edit what the rationale cites, then compare the effect to a matched control edit.

Why this matters

It connects textual explanations to internal representations without re-training, which is especially relevant to Prof. Chen’s class.

Concrete playground

FEVER + GPT-2 XL: fact verification with evidence passages.

02

EF Pipeline Overview

01

Load Claim

02

Generate Rationale

03

Extract Triples (copula-based heuristic)

04

Pick Target / Control

05

Edit with ROME

06

Record Flips

The EF metric is simply: EF = Flip_target − Flip_control. This difference tells us whether the cited fact is more causally important than a matched control. Triple extraction is heuristic but interpretable — it gives us the subject–relation–object fact to edit via ROME (unified model editing).

03

Engineering Challenges & Solutions

Two Main Blockers

GPT-2 XL + ROME is memory hungry

The model-editing setup quickly exceeds available memory when run at scale.

Unified editing assumed Wikipedia stats & fp32

Defaults in the original pipeline did not match our data source or compute format.

Solutions Implemented

Batching generation across examples

Reduced overhead by grouping inference runs.

Per-edit restore hooks

Avoided full model deep copies by restoring only what each edit changes.

fp16 → fp32 casting during edits, back to fp16 after

Matched the editing routine to the required precision, then returned to the lightweight runtime dtype.

Rebuilt inverse-covariance stats on Wikitext

Recomputed the editing statistics on the correct corpus instead of Wikipedia defaults.

dtype alignment throughout pipeline

Kept representations consistent end-to-end to avoid hidden precision mismatches.

Reproducibility Notes

01

30-example cap per run

Each run was limited to 30 examples, so results were aggregated across multiple seeds with merged logs.

02

Fully reproducible pipeline

Precomputed stats, deterministic seeding, and shell scripts make the full workflow repeatable.

04

Quantitative Results

32%

Flip_target

19%

Flip_control

12.5%

EF

Merged across 88 valid examples, 3 seeds of 30

7%

Seed EF min

Range across seeds

24%

Seed EF max

Range across seeds

Qualitative Case Studies

Target-Only Flip ✓ Win

Example: “YouTube has been ranked by a Utah-based web traffic analysis company” → triple edited → answer flips.

Control edit does **not** flip. EF is working correctly.

Control-Only Flip □ Extractor Miss

Extractor latched onto template text rather than the real fact.

Control edit flips but target doesn't. Motivation for improving triple extraction.

Both / Neither Flip

ROME edits can have non-local effects.

Both target and control flip, or neither does. Shows limits of single-point editing.

All case studies are available in the notebook for manual annotation — supporting Prof. Chen's emphasis on human inspection.

06

Discussion & Next Steps

Takeaway

Rationales are partially causal. EF quantifies it — but variance is high due to heuristic extraction and single-point edits.

Limitations

- Heuristic triple extraction (copula-based)
- Single-edit locality assumption
- Small per-seed batch size (30 examples)
- Reliance on GPT-2 XL

Future Work

- Better IE models for relation extraction
- MEMIT for batched edits
- Extend to StrategyQA / prompted CoT

Project Status

- Pipeline + analysis: ✓ Done
- Writing + potential extensions: In progress

Q&A

Questions & Discussion

Invite questions on EF design, engineering trade-offs, or future extensions.

EF Design

How should we handle multi-hop rationales or longer chains of cited facts?

Engineering Trade-offs

What are the limits of ROME locality, and when does MEMIT become necessary?

Future Extensions

How might EF scale to StrategyQA or prompted chain-of-thought settings?

STATUS

Project Status & Deliverables

Phase 1 — Complete ✓

Pipeline Implementation: Data loading, triple extraction (copula-based), ROME editing wrappers, EF metric computation

Phase 2 — Complete ✓

Evaluation & Analysis: Three seeds (30 examples each), merged log of 88 valid cases, bootstrap confidence intervals, ef_eval.py automation

Phase 3 — Complete ✓

Qualitative Case Studies: Notebook with annotated examples (target-only, control-only, both/neither flip cases), plots

Phase 4 — In Progress

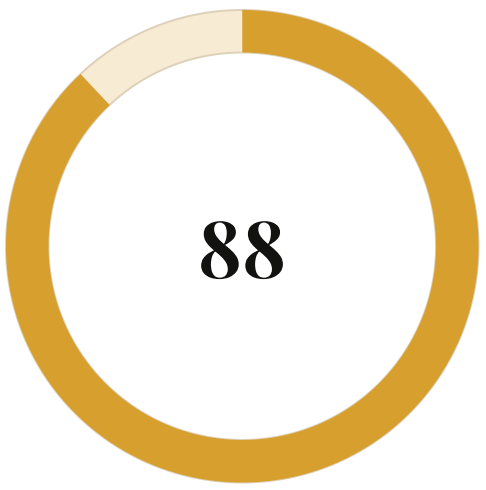
Writing & Extensions: Final write-up in progress; potential extensions to MEMIT batching and StrategyQA

SUMMARY

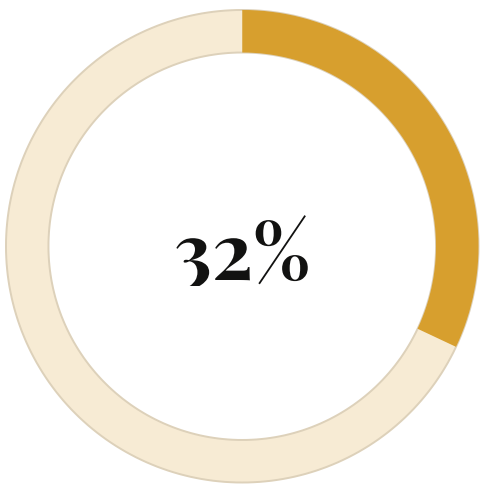
Project Summary

Editability–Faithfulness: Probing the Causal Grounding of Free-Text Rationales via ROME

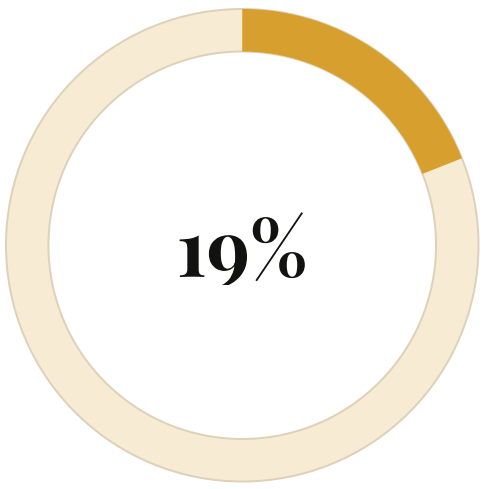
I built an EF evaluation pipeline that measures whether GPT-2 XL's free-text rationales correspond to internal factual knowledge. The system generates FEVER rationales, extracts cited triples, edits those facts with ROME, and checks whether answers flip more often than matched control edits.



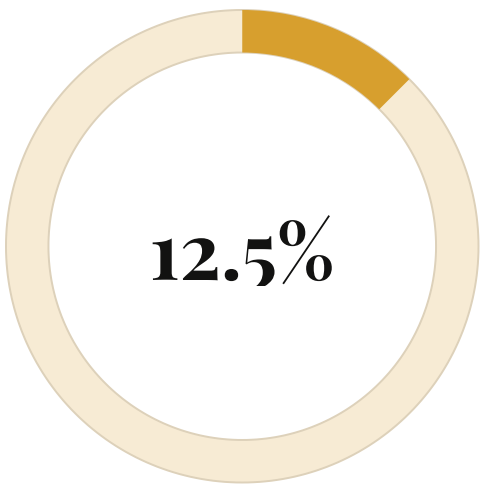
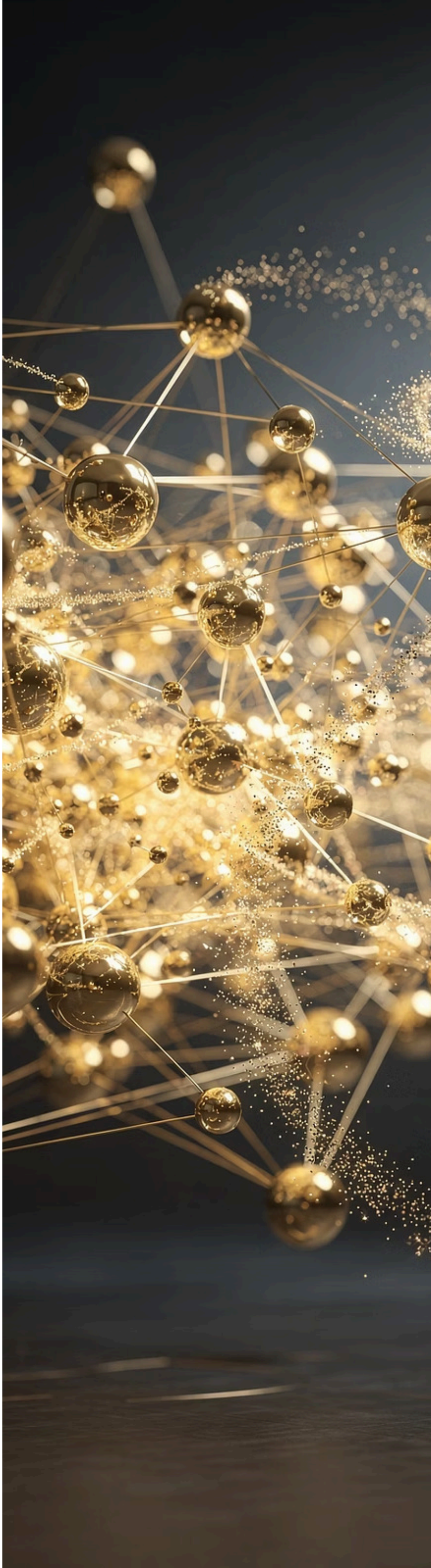
valid cases (3 seeds × 30 examples)



Flip_target ≈ 32%



Flip_control ≈ 19%



EF ≈ 12.5%